

SE1 Module 4 Reviewer: System Models and Requirements

CS0025 Software Engineering 1, Module 4. Built from CS0025-M4S0-1, M4S1, and M4S2. Choice, matching, and typed lists.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Multiple Choice and True or False

1. What is a model in systems development?
 - A. A complete copy of an existing system
 - B. A collection of programming codes
 - C. A list of hardware components
 - D. A simplified description of a system

2. What does modeling a system mean?
 - A. Buying hardware
 - B. Identifying its characteristics, states, and behavior using a notation
 - C. Deleting old files
 - D. Writing the final code

3. What does systems modeling use to conceptualize and construct systems?
 - A. Programming languages only
 - B. Financial statements
 - C. Models and diagrams
 - D. Marketing plans

4. Which of these is one of the system modeling types in the module?
 - A. Cost Modeling
 - B. Marketing Modeling
 - C. Hardware Modeling
 - D. Enterprise Modeling

5. Which modeling approach represents the functions, activities, and operations within a system?
- A. Financial modeling
 - B. Enterprise modeling
 - C. Data modeling
 - D. Functional modeling
6. What does a functional modeling perspective concentrate on?
- A. Storing data only
 - B. Describing the dynamic process
 - C. Hardware cost
 - D. Organizational charts
7. In functional modeling, which element illustrates the transformation of input into output?
- A. Flow line
 - B. Process
 - C. External entity
 - D. Data store
8. In functional modeling, which element is data-collection or some material, shown as a rectangle with the right side usually open?
- A. Process
 - B. Store
 - C. Flow
 - D. External entity
9. In functional modeling, which element is the movement of data or material, shown as an arrow?
- A. External entity
 - B. Store
 - C. Flow
 - D. Process

10. In functional modeling, which element is outside the modeled system but interacts with it?

- A. Store
- B. Flow
- C. Process
- D. External entity

11. What does a functional block diagram describe?

- A. Only the database tables
- B. Hardware prices
- C. Employee records
- D. The functions and interrelationships of a system

12. What does a Functional Flow Block Diagram (FFBD) primarily present?

- A. Hardware prices
- B. Time-sequenced functional activities
- C. Employee records
- D. Database table relationships

13. What is system architecture?

- A. A list of users who can access a system
- B. A financial plan for system development
- C. A conceptual model of a system's structure and behavior
- D. A collection of programming errors

14. What can a system architecture consist of?

- A. Only the user interface
- B. Only hardware
- C. System components and the sub-systems that work together
- D. Only the database

15. What is the primary purpose of Business Process Modeling (BPM)?

- A. To represent, analyze, and improve organizational processes
- B. To determine employee salaries
- C. To identify computer specifications
- D. To install software applications

16. What is a common business objective of BPM?

- A. Increase process speed, increase quality, or reduce costs
- B. Increase cycle time
- C. Reduce customers
- D. Increase paperwork

17. What is a business process?

- A. A programming language
- B. A list of hardware
- C. A random set of tasks
- D. A collection of related, structured activities to produce a specific service or product for a customer

18. Which type of business process creates the organization's primary value stream?

- A. Operational process
- B. Supporting process
- C. Documentation process
- D. Management process

19. Which type of business process governs system operation, including corporate governance and strategic management?

- A. Supporting process
- B. Operational process
- C. Documentation process
- D. Management process

20. Which of the following is an example of a supporting process?

- A. Marketing and sales
- B. Recruitment and technical support
- C. Strategic management
- D. Product manufacturing

21. What is the purpose of Business Process Model and Notation (BPMN)?

- A. To identify computer hardware
- B. To calculate software costs
- C. To provide a graphical notation for business processes
- D. To create database records

22. In BPMN, business processes are specified in which diagram?

- A. Data Flow Diagram
- B. Entity Diagram
- C. Business Process Diagram (BPD)
- D. Gantt Chart

23. What is a business model?

- A. A programming pattern
- B. A type of database
- C. A network diagram
- D. A framework for creating economic, social, or other forms of value

24. What is enterprise modeling?

- A. Buying software
- B. Modeling only a single function
- C. Writing code for enterprises
- D. Representing the structure, processes, information, and resources of a large organization

25. Data modeling is used for what?

- A. Writing test cases
- B. Defining business requirements for a database
- C. Scheduling sprints
- D. Designing hardware

26. Which of the following is normally part of a data model?

- A. Story cards
- B. Entity types, attributes, relationships, integrity rules, and definitions
- C. Sprint backlogs
- D. Only screenshots

27. Which of these is a design diagram named in the module?
- A. Backlog chart
 - B. Sprint diagram
 - C. Data flow diagram
 - D. Story cardboard
28. Which purpose of a design diagram means it is a convenient way to communicate ideas among engineers, designers, and management?
- A. It serves as a communications tool
 - B. It facilitates troubleshooting
 - C. It serves as a planning tool
 - D. It documents the system
29. Which purpose of a design diagram means it acts as a blueprint of the inner workings when diagnosing a failure?
- A. It facilitates troubleshooting
 - B. It documents the system
 - C. It provides an overview
 - D. It defines roles
30. What does a data flow diagram (DFD) illustrate?
- A. Employee schedules
 - B. Source code
 - C. The flow of data and logic within a system
 - D. Hardware cabling only
31. In a DFD, which symbol denotes input and output elements such as a person or place?
- A. Rectangle (entity)
 - B. Rounded rectangle
 - C. Arrow
 - D. Open rectangle

32. In a DFD, activities and operations are represented by what?

- A. A diamond
- B. A rounded rectangle (sometimes a circle), sequentially numbered
- C. An arrow
- D. A rectangle with an open end

33. In a DFD, data flow is represented by what?

- A. An arrow containing a data object
- B. A circle
- C. An open rectangle
- D. A rectangle

34. In a DFD, data stores are represented by what?

- A. A circle
- B. An open rectangle
- C. An arrow
- D. A rounded rectangle

35. How is a manual data store shown in a DFD?

- A. An arrow
- B. A rounded rectangle
- C. A circle
- D. A simple rectangle with an open end

36. How is an automated data store shown in a DFD?

- A. A rounded rectangle
- B. A diamond
- C. A circle
- D. An open rectangle with an additional vertical line near the closed end

37. Which DFD guideline is correct?

- A. Leave symbols unlabeled
- B. Do not mix levels of detail on one chart
- C. Mix all levels in one chart
- D. Use both notations at once

38. Which pair of notations can be used for DFDs?

- A. ER and IDEF
- B. Gane & Sarson or Yourdon & DeMarco
- C. UML and BPMN
- D. Chen and Crow's Foot

39. Which DFD guideline says to check that every symbol is logically connected?

- A. Use only one symbol
- B. Desk check all data flow diagrams
- C. Use a template
- D. Label the top

40. What should be labeled at the top of each data flow diagram?

- A. Nothing
- B. System name, date prepared, and the name of the preparer
- C. Only the date
- D. Only the customer

41. What are system requirements?

- A. A list of users
- B. A legal document
- C. The required specifications a device must have to use certain hardware or software
- D. Only the price of software

42. Most software defines which two sets of system requirements?

- A. Minimum and recommended
- B. Basic and advanced
- C. Free and paid
- D. Old and new

43. Which system requirement is not flexible?

- A. CPU speed
- B. GPU
- C. The operating system and disk space needed for installation
- D. RAM

44. What does requirements analysis encompass?

- A. Only maintenance
- B. Only testing
- C. The tasks of investigating, scoping, and defining a new or altered system
- D. Only writing code

45. What is the first activity of requirements analysis?

- A. The preliminary investigation
- B. Testing
- C. Deployment
- D. Coding

46. Which technique can analysts use to collect information about the current system from potential users, but is time consuming?

- A. Observation
- B. Research
- C. Interviews
- D. Record inspections

47. Which fact finding technique uses reports, bills, policy manuals, regulations, and standard operating procedures?

- A. Questionnaires
- B. Record inspections or reviews
- C. Interviews
- D. Observation

48. Which fact finding technique collects data from large groups?

- A. Observation
- B. Research
- C. Questionnaires
- D. Interviews

49. Which questionnaire type is qualitative and answered in the respondent's own words?
- A. Multiple choice
 - B. Yes or no
 - C. Closed-ended
 - D. Open-ended
50. Which questionnaire type uses prescribed answers and generates quantitative results?
- A. Oral
 - B. Essay
 - C. Open-ended
 - D. Closed-ended
51. Which fact finding technique is a skill where analysts identify the right information, person, and place?
- A. Interviews
 - B. Research
 - C. Questionnaires
 - D. Observation
52. Which advanced fact finding technique obtains information from published materials to verify facts?
- A. Interviews
 - B. Research
 - C. Questionnaires
 - D. Observation
53. An operating system is a system requirement for a computer.
- A. False
 - B. True
54. A design diagram is a communications tool and a planning tool.
- A. False
 - B. True

55. Interviews are quick and cheap for collecting information.

- A. False
- B. True

Matching (Dropdown Style)

56. Match the DFD symbol with its shape.

- A. Rounded rectangle or circle
- B. Diamond
- C. Open rectangle
- D. Arrow
- E. Rectangle

Entity: _____

Process: _____

Data flow: _____

Data store: _____

57. Match the business process type with its example.

- A. Software testing only
- B. Corporate governance and strategic management
- C. Accounting, recruitment, technical support
- D. Purchasing, manufacturing, marketing, sales

Management process: _____

Operational process: _____

Supporting process: _____

58. Match the fact finding technique with its description.

- A. Published materials used to verify facts
- B. Data from large groups; open or closed
- C. Choose the right person and place to watch
- D. Guessing
- E. Reports, bills, policy manuals, procedures
- F. Discover misunderstandings and resistance; time consuming

Interviews: _____

Record inspections: _____

Questionnaires: _____

Observation: _____

Research: _____

59. Match the modeling type with its description.

- A. Functions, activities, and operations of a system
- B. Conceptual model of structure and behavior
- C. Defines business requirements for a database
- D. Modeling of hardware prices
- E. Represents processes so they can be analyzed and improved

Functional modeling: _____

System architecture: _____

Business process modeling: _____

Data modeling: _____

60. Match the questionnaire type with its property.

- A. Quantitative; prescribed answers
- B. Oral only
- C. Qualitative; answers in own words

Open-ended: _____

Closed-ended: _____

Typed Lists (Enumeration)

61. List the four types of system modeling in the module. (4 items)

1. _____
2. _____
3. _____
4. _____

62. List the four elements of a functional model. (4 items)

1. _____
2. _____
3. _____
4. _____

63. List the three types of business process. (3 items)

1. _____
2. _____
3. _____

64. List the four DFD symbols. (4 items)

1. _____
2. _____
3. _____
4. _____

65. List the five fact finding techniques. (5 items)

1. _____
2. _____
3. _____
4. _____
5. _____

66. List the two types of questionnaires. (2 items)

1. _____
2. _____

67. List the two sets of system requirements. (2 items)

1. _____
2. _____

68. List any four design diagrams named in the module. (4 items)

1. _____
2. _____
3. _____
4. _____

69. List any four purposes of a design diagram. (4 items)

1. _____
2. _____
3. _____
4. _____

70. List the two DFD notations. (2 items)

1. _____
2. _____

71. List the four components of a data model. (4 items)

1. _____
2. _____
3. _____
4. _____

Answer Key

No.	Answer	Why
1	D. A simplified description of a system	Ch. 1: System models
2	B. Identifying its characteristics, states, and behavior using a notation	Ch. 1: System models
3	C. Models and diagrams	Ch. 1: System models
4	D. Enterprise Modeling	Ch. 1: System models
5	D. Functional modeling	Ch. 1: Functional modeling
6	B. Describing the dynamic process	Ch. 1: Functional modeling
7	B. Process	Ch. 1: Functional modeling
8	B. Store	Ch. 1: Functional modeling
9	C. Flow	Ch. 1: Functional modeling
10	D. External entity	Ch. 1: Functional modeling
11	D. The functions and interrelationships of a system	Ch. 1: Functional block diagram
12	B. Time-sequenced functional activities	Ch. 1: FFBD
13	C. A conceptual model of a system's structure and behavior	Ch. 1: System architecture
14	C. System components and the sub-systems that work together	Ch. 1: System architecture
15	A. To represent, analyze, and improve organizational processes	Ch. 1: BPM
16	A. Increase process speed, increase quality, or reduce costs	Ch. 1: BPM
17	D. A collection of related, structured activities to produce a specific service or product for a customer	Ch. 1: Business process
18	A. Operational process	Ch. 1: Business process types
19	D. Management process	Ch. 1: Business process types
20	B. Recruitment and technical support	Ch. 1: Business process types
21	C. To provide a graphical notation for business processes	Ch. 1: BPMN
22	C. Business Process Diagram (BPD)	Ch. 1: BPMN
23	D. A framework for creating economic, social, or other forms of value	Ch. 1: Business model
24	D. Representing the structure, processes, information, and resources of a large organization	Ch. 1: Enterprise modeling
25	B. Defining business requirements for a database	Ch. 1: Data modeling
26	B. Entity types, attributes,	Ch. 1: Data modeling

	relationships, integrity rules, and definitions	
27	C. Data flow diagram	Ch. 1: Design diagrams
28	A. It serves as a communications tool	Ch. 1: Design diagram purposes
29	A. It facilitates troubleshooting	Ch. 1: Design diagram purposes
30	C. The flow of data and logic within a system	Ch. 1: DFD
31	A. Rectangle (entity)	Ch. 1: DFD symbols
32	B. A rounded rectangle (sometimes a circle), sequentially numbered	Ch. 1: DFD symbols
33	A. An arrow containing a data object	Ch. 1: DFD symbols
34	B. An open rectangle	Ch. 1: DFD symbols
35	D. A simple rectangle with an open end	Ch. 1: DFD symbols
36	D. An open rectangle with an additional vertical line near the closed end	Ch. 1: DFD symbols
37	B. Do not mix levels of detail on one chart	Ch. 1: DFD guidelines
38	B. Gane & Sarson or Yourdon & DeMarco	Ch. 1: DFD guidelines
39	B. Desk check all data flow diagrams	Ch. 1: DFD guidelines
40	B. System name, date prepared, and the name of the preparer	Ch. 1: DFD guidelines
41	C. The required specifications a device must have to use certain hardware or software	Ch. 2: System requirements
42	A. Minimum and recommended	Ch. 2: System requirements
43	C. The operating system and disk space needed for installation	Ch. 2: System requirements
44	C. The tasks of investigating, scoping, and defining a new or altered system	Ch. 2: Requirements analysis
45	A. The preliminary investigation	Ch. 2: Requirements analysis
46	C. Interviews	Ch. 2: Fact finding
47	B. Record inspections or reviews	Ch. 2: Fact finding
48	C. Questionnaires	Ch. 2: Fact finding
49	D. Open-ended	Ch. 2: Questionnaires
50	D. Closed-ended	Ch. 2: Questionnaires
51	D. Observation	Ch. 2: Fact finding
52	B. Research	Ch. 2: Fact finding
53	B. True	Ch. 2: System requirements

54	B. True	Ch. 1: Design diagram purposes
55	A. False	Ch. 2: Interviews are time consuming
56	Entity = E; Process = A; Data flow = D; Data store = C	Ch. 1: DFD symbols
57	Management process = B; Operational process = D; Supporting process = C	Ch. 1: Business process types
58	Interviews = F; Record inspections = E; Questionnaires = B; Observation = C; Research = A	Ch. 2: Fact finding
59	Functional modeling = A; System architecture = B; Business process modeling = E; Data modeling = C	Ch. 1
60	Open-ended = C; Closed-ended = A	Ch. 2: Questionnaires
61	Functional Modeling; System Architecture; Business Process Modeling; Enterprise Modeling (in order)	Ch. 1: System models
62	Process, Store, Flow, External Entity (any order)	Ch. 1: Functional modeling
63	Management Process; Operational Process; Supporting Process (in order)	Ch. 1: Business process types
64	Entity (rectangle), Process (rounded rectangle or circle), Data flow (arrow), Data store (open rectangle) (any order)	Ch. 1: DFD symbols
65	Interviews, Record inspections or reviews, Questionnaires, Observation, Research (any order)	Ch. 2: Fact finding
66	Open-ended, Closed-ended (any order)	Ch. 2: Questionnaires
67	Minimum, Recommended (any order)	Ch. 2: System requirements
68	Data flow diagrams, Structured charts, Decision trees, Block diagrams (cycle of operations) (any order)	Ch. 1: Design diagrams
69	Communications tool, Planning tool, Provides an overview of the system, Defines roles (any order)	Ch. 1: Design diagram purposes (eight in total: also demonstrates relationships, promotes logical procedures, facilitates troubleshooting, documents the system)
70	Gane & Sarson, Yourdon & DeMarco (any order)	Ch. 1: DFD guidelines
71	Entity types, Attributes, Relationships, Integrity rules (any	Ch. 1: Data modeling

	order)	
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